

SUPPLEMENTARY FILE

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Paper under double-blind review

1 Effectiveness of Contrastive Losses on other pre-trained models

Table 1: Results on different Loss function and neighbor sample algorithm on RoBERTa model.

Methods		Java		Python		PHP		JS		Ruby		Go	
		MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1
bi-kNN	Full(MNNCoCo)	0.692	0.599	0.682	0.583	0.650	0.549	0.594	0.498	0.637	0.540	0.895	0.851
	-w/o MCL	0.684	0.592	0.680	0.579	0.646	0.545	0.590	0.493	0.636	0.540	0.890	0.846
uni-kNN	Full	0.683	0.592	0.678	0.571	0.640	0.542	0.585	0.489	0.632	0.535	0.886	0.843
	-w/o MCL	0.662	0.573	0.678	0.572	0.637	0.541	0.585	0.490	0.632	0.536	0.886	0.843
-w/o NNCL		0.624	0.535	0.610	0.512	0.587	0.503	0.528	0.435	0.598	0.502	0.867	0.816
-w/o all		0.605	0.504	0.590	0.499	0.561	0.464	0.523	0.427	0.587	0.473	0.855	0.802

Table 2: Results on different Loss function and neighbor sample algorithm on RoBERTa (code) model.

Methods		Java		Python		PHP		JS		Ruby		Go	
		MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1
bi-kNN	Full(MNNCoCo)	0.732	0.644	0.727	0.632	0.683	0.583	0.669	0.572	0.708	0.614	0.913	0.873
	-w/o MCL	0.728	0.640	0.724	0.632	0.680	0.579	0.661	0.568	0.702	0.612	0.908	0.870
uni-kNN	Full	0.727	0.636	0.718	0.630	0.674	0.577	0.653	0.562	0.700	0.607	0.906	0.869
	-w/o MCL	0.708	0.609	0.719	0.631	0.672	0.574	0.654	0.565	0.7001	0.608	0.901	0.862
-w/o NNCL		0.648	0.554	0.642	0.545	0.593	0.510	0.578	0.498	0.639	0.537	0.573	0.816
-w/o all		0.636	0.537	0.621	0.533	0.581	0.490	0.570	0.487	0.631	0.526	0.864	0.806

Table 3: Results on different Loss function and neighbor sample algorithm on GraphCodeBert model.

Methods		Java		Python		PHP		JS		Ruby		Go	
		MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1	MRR	P@1
bi-kNN	Full(MNNCoCo)	0.739	0.653	0.739	0.648	0.691	0.593	0.687	0.597	0.735	0.643	0.919	0.880
	-w/o MCL	0.734	0.647	0.740	0.647	0.687	0.590	0.674	0.586	0.736	0.645	0.913	0.872
uni-kNN	Full	0.732	0.644	0.736	0.641	0.682	0.586	0.669	0.564	0.731	0.624	0.913	0.872
	-w/o MCL	0.728	0.619	0.736	0.640	0.679	0.583	0.669	0.565	0.730	0.623	0.910	0.868
-w/o NNCL		0.701	0.602	0.711	0.615	0.653	0.554	0.639	0.539	0.708	0.589	0.902	0.847
-w/o all		0.691	0.596	0.692	0.594	0.649	0.549	0.644	0.546	0.703	0.581	0.897	0.843